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Roll No

FT-605 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Nuclear Safety and Radioactive Materials
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Name the different method for the dosimetric measurements.
2. Describe the facilities for cold and hot radiological laboratories.
3. Explain the principle of heat detector for fire-fighting and rescue operations.
4. Describe concentration, treatment and disposal techniques of radioactive liquid effluents.
5. a) What are biological effects of Radiation?
b) Briefly explain radiation dosimetry.
6. Explain the absorption of gaseous waste in organic solvents.
7. How quality assurance is important in nuclear power plant from safety point of view?

OR

What are the three safety objectives defined for nuclear power plant? Explain them.

8. Answer any three of the following:
 - a) Discuss the calculation of absorbed dose in media from ionization chamber using exposure dose in air.
 - b) Describe the materials of construction for radiological laboratories.
 - c) What is Radioactive emission.
 - d) What is controlled zone.
 - e) Explain nuclear explosion.
 - f) Explain safety requirements in a nuclear power plant.
